

AGRICULTURAL EXTENSION

The use of the term 'extension' originated in England in 1866 with a system of university extension which was taken up first by Cambridge and Oxford Universities, and later by other educational institutions in England and in other countries.

Term 'extension education' used first in 1873 by Cambridge University, to carry the educational advantages of universities to ordinary people.

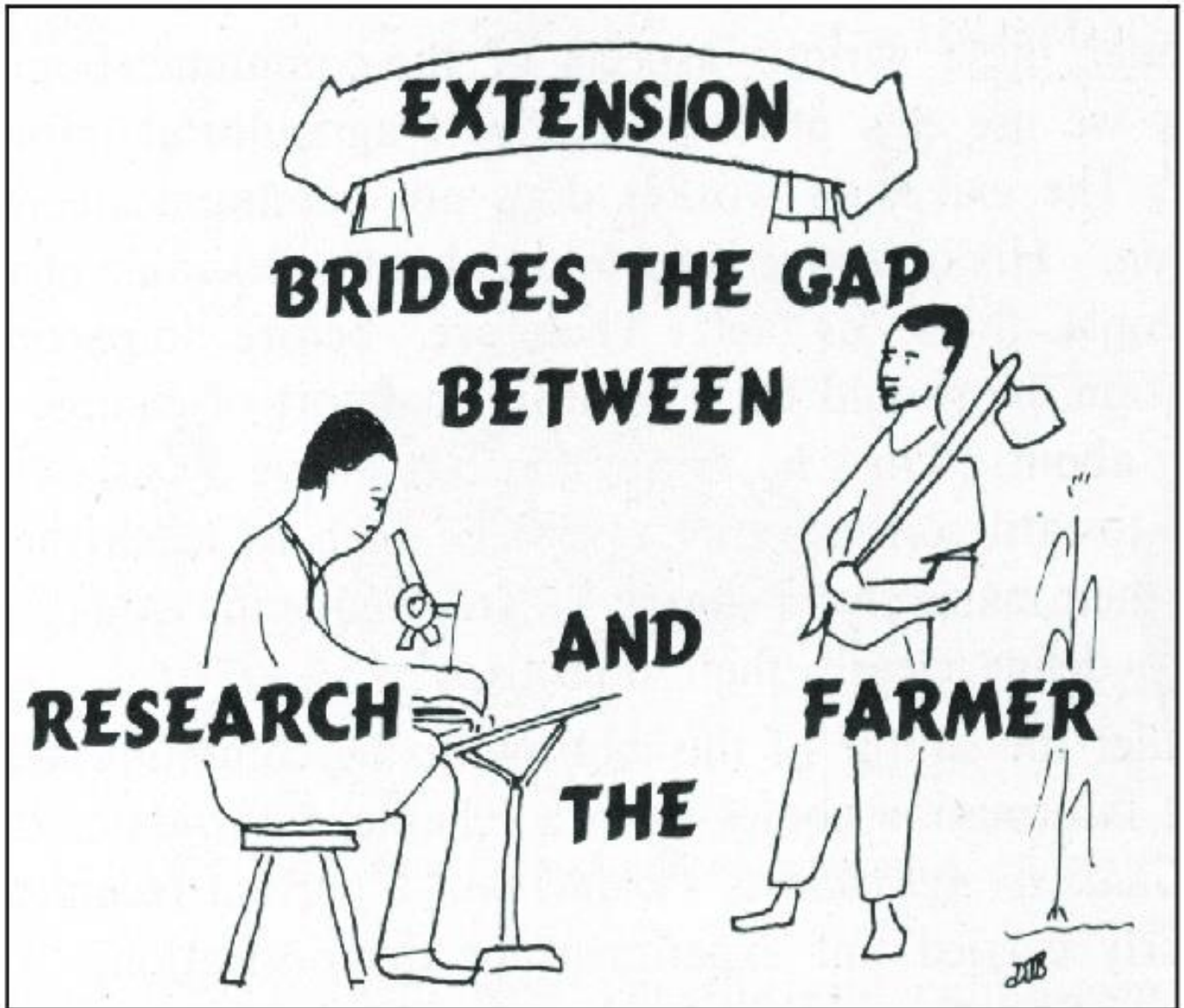


Historically, **extension has meant education in agriculture and in home economics for rural people.** This education is practical, aimed at improving farm and home.

Leagans (1961) conceptualized **extension education as an applied science consisting of content derived from research, accumulated field experiences and relevant principles drawn from the behavioural sciences** synthesized with useful technology into a body of philosophy, principles, content and methods focused on the problems of out-of-school education for adults and youth.

Extension may be defined as the science of developing capability of the people for sustainable improvement in their quality of life.

Agricultural extension is a general term meaning the application of scientific research and new knowledge to agricultural practices through farmer education. The field of extension now encompasses a wider range of communication and learning activities organized for rural people by professionals from different disciplines, including agriculture, agricultural marketing, health, and business studies.



The Objective/Importance of Agricultural Extension:

- **To educate farmers about farming innovations** and prepare them for change and development
- **To provide improved technology, information and skills** to farmers for increasing agricultural and livestock production, and the productivity;
- **To give feedback from the field to the research stations** on the application of technology generated by them;

- To **conduct demonstrations in farmers' fields on** the potentials of improved farm technology;
- **To assist farmers to get access to services and supplies**, credit and marketing; and
- **To help them develop resources like land and water.**

Extension Education – The extension education role is **generally performed by the higher learning institutions like the Agricultural and other Universities and Colleges, ICAR Institutes**, Home Science Colleges and apex level Training and Extension Organizations. At the university level, extension is integrated with teaching and research, while at the research institutes, extension is integrated with research. **At the other apex level organizations, extension is generally integrated with training in extension.**

Extension service – It is mainly to provide educational service to the people according to their need, for improving their life through better working. The main responsibility of extension service is with the State Government. **The departments of Agriculture, Horticulture, Animal Husbandry, Veterinary, Forestry, Fishery, Sericulture, etc. of the State Government carry out extension work with the farmers and rural people over the entire State.** The departments maintain close contact with the relevant Universities and Research Institutes for obtaining appropriate technology and methodology for extension work, and for providing them with feedback information from the field for research.

Principal Of Extension Education

1. Principle Of Peoples Need and Interest

Always programme must be **developed according needs & interests** this need differ **from individual to individual, from village to village, from block to block**, from state to state; therefore, there cannot be one programme for all people

2. Principle of grass root level organisation

A group of rural people in local community should sponsor extension work. They work with local community so that the programme should fit in with the local conditions. The **aim of organising the local group is to demonstrate the value of the new practices** or programmes so that more & more people would participate

3. Principle of cultural difference

Extension work is **based on the cultural background of the people with whom the work is done**. Differences in the culture are always being there between extension worker and rural people, success is **when extension professionals have to know the level of the knowledge, & the skills of the people, methods & tools used by them**, their customs, traditions, beliefs, values, etc. before starting the extension programme.

4. Principle of cooperation and people's participation

Extension is a co-operative venture. **It is a joint democratic enterprise in which rural people cooperate with their village, block & state officials to pursue a common cause. Ultimately without the cooperation of people the work cannot be successful and desired result cannot be achieved.** The first task of extension education is the cooperation of people and their participation in work. Extension helps people to help themselves.

5. Principle of cultural change

Extension education starts with what the learner knows, has and thinks. With this in mind and with an attitude of respect towards **clients, the extension professionals must seek to discover and understand the limitations, taboo and the cultural values related to each phase of programme** so that an acceptable approach could be selected in the locality.

6. Principle of learning by doing

According to this principle, **farmers are encouraged to learn by doing the work themselves and by participating in it**. When a person does a work, **he gains practical knowledge and experiences the difficulties**. Extension professionals are able to understand the problems and provide proper guidance to the farmers and thus, they are able to receive proper information/feedback

7. Principle of trained specialists

It is very difficult that **extension personnel should be knowledgeable about all problems**. Therefore, it is necessary that specialists should impart training to the farmers from time to time.

8. Principle of adaptability in use of extension teaching methods

People differ from each other; one group differs from another group and conditions also differ from place to place. An extension **programme should be flexible, so that necessary**

changes can be made whenever needed, to meet the varying conditions. Extension professionals should have knowledge of extension methods so that they can select proper method according to the condition. **Teaching methods should be flexible so that they can be properly applied on people according to their age groups, educational background, economic standard and gender.** In extension education, two or more methods should be applied according to the principle of adaptability

9. Principle of leadership

Extension work is based on the full utilisation of local leadership. **The selection & training of local leaders to enable them to help in carrying out extension work is essential to the success of the programme.** People have more faith in local leaders & they should be used to put across a new idea so that it is accepted with the least resistance.

10. Principle of whole family

Extension work **will have a better chance of success if the extension professionals have a whole-family approach instead of piecemeal approach or separate & uninterested approach.** **Extension work is, therefore, for the whole family, i.e. for male, female and children.**

11. Principle of evaluation

Extension is **based upon the methods of science, and it needs constant evaluation.** The effectiveness of the work is measured in terms of the **changes brought about in the knowledge, skill, attitude, and adoption behaviour of the people**, not merely in terms of achievement of physical targets

12. Principle of satisfaction

The **end-product of the effort of extension teaching is the satisfaction that comes to the farmer and his family members** as the result of solving a problem, **meeting a need, acquiring a new skill or some other changes in behaviour.** Satisfaction is the key to success in extension work. A satisfied stakeholder is the best advertisement.

13. Principal of Indigenous Knowledge

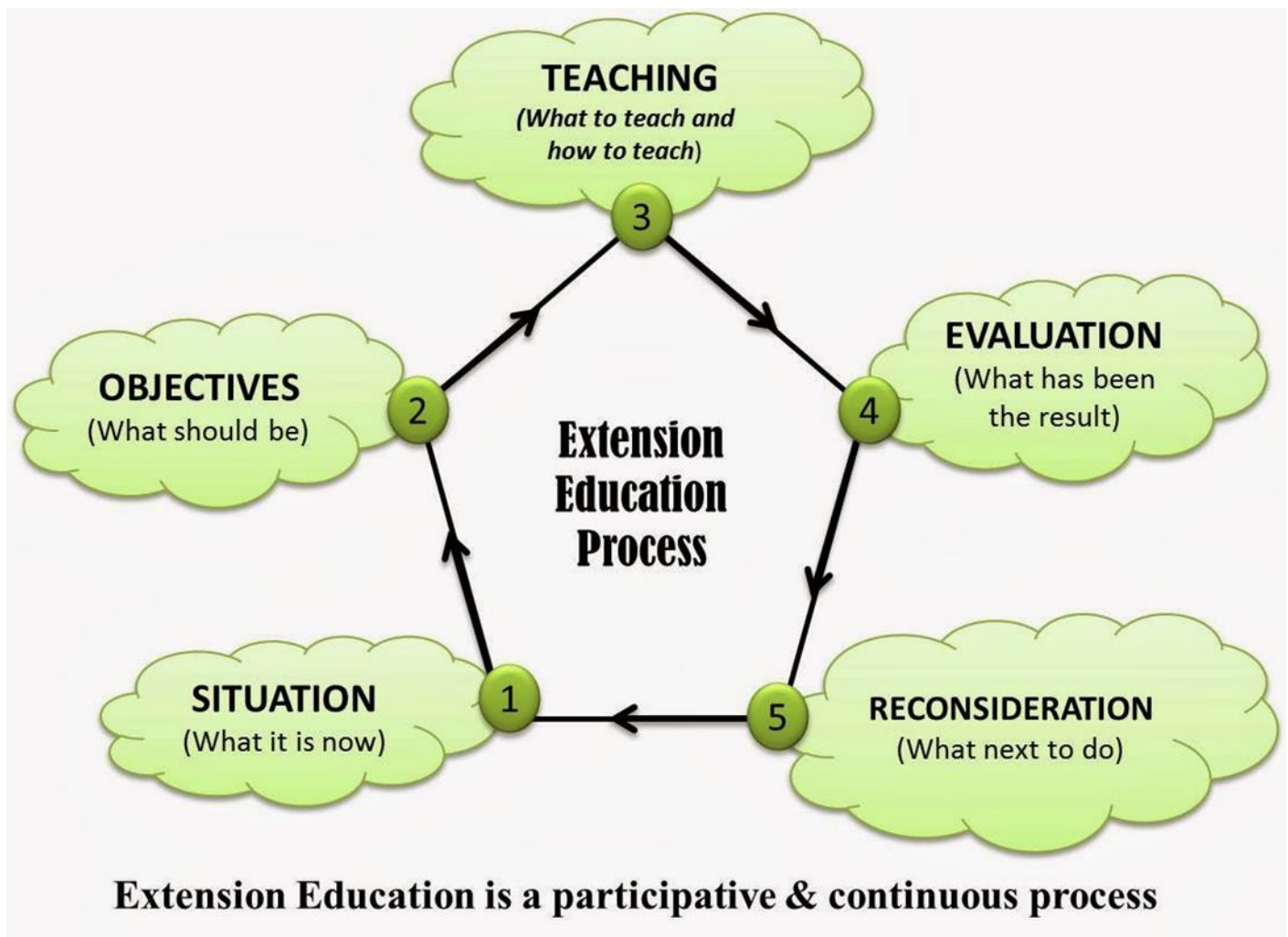
People everywhere **have indigenous knowledge system which they have develop through generation of work experience and problem solving** in their own specific

situation. **The indigenous knowledge systems encompass all aspect of life and people considered it essential for their survival**

PROCESS AND STEPS INVOLVED IN AGRICULTURAL EXTENSION

First Step: The first step consists of **collection of facts and analysis of the situation**. Facts about the people and their enterprises: **the economic, social, cultural, physical and technological environment in which they live and work**. These may be obtained by appropriate survey and establishing rapport with the people.

Second Step: The next step is deciding on realistic objectives which may be accomplished by the community. **A limited number of objectives should be selected by involving the local people. The objectives should be specific and clearly stated, and on completion, should bring satisfaction to the community**. Objectives should state the behavioural changes in people as well as desired economic and social outcomes.



Third Step: The third step is teaching, **which involves choosing what should be taught (the content) and how the people should be taught** (the methods and aids to be used). It requires **selecting research findings of economic and practical** importance relevant to the community, and selection and combination of appropriate teaching methods and aids.

Fourth Step: The fourth step is **evaluating the teaching**, i.e. determining the extent to which the objectives have been reached. **To evaluate the results of an educational programme objectively, it is desirable to conduct a re-survey.** The evidence of changed behaviour should be collected, which shall not only provide a measure of success, but shall also indicate the deficiencies, if any.

Fifth Step: The fifth step is **re-consideration of the entire extension educational programme on the light of the results of evaluation.** The problems identified in the process of evaluation may become the starting point for the next phase of the extension educational programme, unless new problems have developed or new situations have arisen.

Monitoring and Evaluation of Extension Programme

Evaluation as applied to the field of extension education, may be defined as "a process of systematic appraisal by which we determine the value, worth or meaning of an activity or an enterprise". **It is a method for determining how far an activity has progressed and how much further it should be carried to accomplish objectives.** Thus, to an extension worker **evaluation means determining the results of his extension programmes** in order to know the extent to which objectives have been achieved and why and what changes would be needed in case the programme is planned again, or in its implementation

accomplished in line with the objectives and also to determine the reasons for success or failure

1	2	3	4	5
Casual everyday evaluation	Self-checking evaluation	Do-it yourself evaluation	Extension evaluation studies	Scientific research

Informal**Formal**

Types Of Evaluation

1. Self-evaluation

This is to be carried out by every worker as a matter of routine. **This requires the self-critical attitude which is so essential for extension work.** By this self-critical attitude, the chances of an extension worker growing and continuously improving his professional competency become greater

2. Internal evaluation:

Evaluation **carried to by the agency responsible for the planning** and implementation of the programme.

Some of the other methods for internal evaluation are:

- **Systematic use of diaries** and reports of workers
- **Planned visits of staff members** to work spot
- **Use of special questionnaires** and proforma for observation and inquiry etc.

3. External evaluation:

Evaluation conducted by a person or a committee outside the area of operation. One of the strong features of the **Indian Community Development Programme is that simultaneous with its start an independent agency, namely the Programme Evaluation Organization, was established.**

Steps involved in an extension programme evaluative process may be as follows

I. Formulate evaluation objectives

Specific objectives to be **achieved through the evaluative process must be clearly and adequately identified and started.** All further efforts should be knit around these objectives.

II. Classify programme objectives

It is assumed **that each extension programme, when formulated and implemented, will have specific well-defined objectives.** Since evaluation is basically a process of determining the extent to which various extension teaching activities were organized and managed and the **extent to which they contributed to achieving the goals, programme**

objectives must be clearly understood and if necessary, further broke down into measurable terms.

This is a crucial step as all further efforts will be directed towards collecting evidence related to these objectives.

1. Identify indicators

To identify indicators or the kind of evidence necessary to evaluate achievement in relation to specified programme objectives, **it is necessary that specific beneficiaries of the programme be identified, the kind of behavioural changes expected in them be clearly stated, and the kin of learning experiences** expected to be provided to them spelled out, together with the level of management to be achieved for provided those learning experiences are specified. Once this is done, identification of specific indicators to measures the achievements will not be difficult.

2. Decide the kind of information needed

Once the indicators **for evaluating the management and performance of a programme have been indicated**, specific information to be collected may be worked out. Since there is **usually more information than an extension worker can collect**, he has to be very discriminating about the kind and amount of information that should be collected. **Timing for collection of information may also need to be specified**

3. Sampling

The purpose of sampling is to **take a relatively small number of units from a population in such a way that the evidence collected from them becomes representative evidence of the entire population**. Although there are **several sampling methods, perhaps stratified sampling procedures may be most suitable** for extension evaluation studies they allow **inclusion of all interested groups** and ensure enough heterogeneity in the sample

4. Decide the design of evaluation

An ideal design of evaluation may be an experimental one. This would allow separating the effect of the programme from other factors, by setting control and treatment groups. Several experimental designs, **such as one-group pre-test-post-test design, static groups comparison, pre-test, post-test control group design**, Solomon four-group design, longitudinal study design, etc. are available in literature and can be used.

However, in actual practice, **extension programmes are seldom run in a way that allows an experimental design of evaluation.** In Pilot Projects, it might be possible to use an experimental design of evaluation. **By and large, a survey method is use.** This method can be used for evaluating ongoing progress or as an ex-post facto evaluation of the programme after it has completed its tenure.

5. Collection and analysis of evaluation evidence

There are many methods for collecting information **for evaluative purposes, such as the mail questionnaire, personal interview, distributed questionnaires,** group interviews, case studies, systematic field observations, systematic study of secondary data etc. **Selection of the right kind of data collection method will depend on the objectives of the evaluation, kind of information needed, time** and resources available and the type of respondents from whom information is to be collected

6. Interpretation of the results in a proper way

It is a very crucial as evaluation results can be missed also. **Once tentative generalizations are arrived at, it may be appropriate and they are informally discussed** among the interpreters as well as with **programme planning and implementation officials, so** that the results of evaluation are put in a proper perspective

EXTENSION TEACHING METHODS

In case of Agricultural Extension, the main theme **of teaching happens to be the adoption of innovation by individual farmer and subsequent diffusion of the same in the community.**

Extension workers' main job is to educate rural people. Extension methods are the tools of the extension workers. **These tools are called extension-teaching methods.** **Extension teaching methods may be defined as the devices used to create situations in which communication can take place between the extension worker and learner.**

Effectiveness of the teaching methods depends on the combination of the same. Research studies have conclusively shown that **the adoption of innovation is positively related to the combinations of different extension teaching methods,** at different stages of learning. The choice of a channel or method of communication, also known as extension

teaching method, generally depends on the number and location of the target audience and the time available for communication.

They Are Categorized As

1. Individual method
2. Group method
3. Mass Method

Individual Method In individual method the **extension agent communicates** with the **people individually, maintaining separate identity of each person**. This method is followed when the **number of people to be contacted are few**, are conveniently located close to the communicator, and sufficient time is available for communication.

Some examples of individual method are: Farm and home visit, farmers' call, personal letter, adaptive or mini kit trial and farm clinic.

Individual methods are:

- **Farm and Home Visit:** It is a **direct face-to-face** contact by the extension agent with the farmer or homemaker at their farm or home for extension work
- **Farmers' Call:** It is a **call made by farmer or home maker** at the working place of the extension agent for obtaining information and assistance.
- **Personal Letter:** This **letter is written by the extension agent to particular farmer** or home maker in connection with extension work. This should be regarded as a substitute for personal contact.
- **Adaptive or Mini-kit Trial:** It is a method of determining the suitability or otherwise of a new practice in farmers' situation. **This may be regarded as an on farm participatory technology development practice in which farmer's choice and opinion about the practice are most important.**
- **Farm Clinic:** Farm clinic is a facility developed and extended to the farmers for diagnosis and **treatment of farm problems and to provide** some specialized advice to individual farmers.

Group Method: A group may be defined as an aggregate of small number of people in **reciprocal communication and integration around common interest**. In this method, the **extension agent communicates with the people in groups** and not as individual persons. This method is adopted when it is necessary to communicate with a number of

people simultaneously; **who are located not far off from the communicator and reasonably good time is available for communication.**

The examples of group methods are resulting demonstration, method demonstration, group meeting method, small group training, field day or farmer's day and study tour or exposure visit.

The group methods are:

- **Result Demonstration:** It is a *method of motivating the people for adoption of a new practice by showing its distinctly superior result.* The demonstrations are conducted in the farm or home of selected individuals and are utilized to educate and motivate groups of people in their neighbourhood. **This is a very effective method for the transfer of technology in a community**
- **Method Demonstration:** It is relatively short-time demonstrations given before a group of people to show how to carry out entirely new practice or a old practice in a better way. **It is essentially skill training, where the emphasis is on effectively carrying out a job,** which shall improve upon the result.
- **Group Meeting:** It is a method of **democratically arriving at certain decisions by a group of people, by taking into consideration the members' point of view.** Group meetings and discussions aim at collective decision making and at improving individual decision making by using the knowledge and experience of group members.
- **Small Group Training:** It is a technique of **imparting specific skills to a group of people who need them by creating appropriate learning situation.** This is an effective method for transfer of technology.
- **Field Day Or Farmer's Day:** A method of influencing the people to adopt a new practice by showing what has actually been achieved by applying the practice under field conditions. **A field day or farmers' day may be held in a research farm or in a farmer's field or home.**
- **Study tour or Exposure visits:** In study tour, **a group of interested persons accompanied and guided by one or more extension agents moves** out of their neighbourhood to study and learn significant improvements in farm and home elsewhere.

Mass Contact Method:

In this method, the extension agent communicates with a vast and heterogeneous mass of people, without taking into consideration their individual or group identity. **This method is valid when a large and widely dispersed audience is to be communicated within a short time.** There may be a few communicators such as the extension agent and some subject matter specialists. **The size of the audience may be a few 100s in mass meeting, few 1000s in campaign and a few lakhs in newspaper, radio and television.**

A few examples of mass methods are farm publications, mass meeting, campaign, exhibition, newspaper, radio and television.

Mass Methods Are:

- **Farm Publication:** It is a **type of publication prepared by the extension agency in printed form, containing information relating to the improvement of farm and home.**

Farm publications **are of various types such as leaflet, folder, bulletin, newsletter, journal and magazine.** Farm publications may be used singly or in combination with other extension methods.

- **Mass meeting:** It is held to **communicate interesting and useful information to a large audience at a time.** The size of the **audience for mass meeting may be a few hundreds, but at the time of fairs or festivals,** it may be few thousands.
- **Campaign:** It is an **intensive educational activity for motivating and mobilizing a community to action to solve a problem or specify a need urgently felt by it.** **The duration of a campaign may be for a single day on a theme like 'Water for Life',** for a few weeks as in **Rat control**, for a few months as in Vanamahotsava (tree planting) and for a few years as in 'Grow More Food' campaign.
- **Exhibition:** It is a systematic display **of models, specimens, charts, photographs, pictures, posters, information etc. in a sequence around a theme to create awareness and interest in the community.**

This method is suitable for reaching all types of people. **Exhibitions may be held at the village, block, sub-division, district, State, national and international levels.**

- **Newspaper:** It is **a bunch of loose printed papers properly folded**, which contains news, views, events, advertisements etc. and is offered for sale at regular intervals particularly daily or weekly. **Newspapers are usually printed on a special type paper, known as newsprint.**
- **Radio:** It is an **electronic audio medium for broadcasting programmes** to the audience. **This medium is suitable for communication to millions of people widely dispersed and suited in far-flung remote areas.** Availability of low-cost transistor sets has helped radio to penetrate deep into the rural life
- **Television:** It is an **electronic audio-visual medium which provides pictures with synchronized sound.** Television combines immediacy of radio with the mobility of cinema and can carry messages over long distances at all relatively low unit cost. It is cosmopolite in approach and can be used to create instant mass awareness.

Examples of different methods

Individual	Group (2-30 persons)	Mass or community (more than 30 person)
Farm & home visit or personal contact	Symposium (2-5 speaker)	Bulletins
Letters (Personal)	Pannel (2-8 speakers)	Leaflet/booklet/folder/pamphlet
Telephone call	Discussion	Newspaper
Office call	Lecture	Magazine
Result demonstration	Tour and field days	Journal
	Method demonstration	Exhibition
	Meeting	Television
	Role playing	Radio
	Puppet show	Circular letter
	Slide show	Fairs
	PAE set (Public address set)	Posters
	School	Cinema (Film)
	Conference	Drama
	Seminar	Songs
	Training camp	
	Flash card	

Demonstration

Demonstration means showing by doing. The basic principle of demonstration is learning by seeing and doing.

There are three types of demonstration: -

a. Method Demonstration

- It is a **short type demonstration**
- Its **main purpose is to provide only skill**
- It **does not compare between the old and new technique or skill.**
- It is the **oldest form of teaching**
- The father of **method demonstration was Dr. seeman. A. Knapp.**
- It is a **single Practice demonstration** i.e how to operate a tractor, how to apply fertilizer in the field.
- In this demonstration **any process is shown and made clear to the people by doing in a sequence starting from the beginning to the end** and zealous person is given opportunity for doing

b. Result demonstration

- It is a **long type** of demonstration
- It shows the **value or worth** of the new practice
- **Comparison between two particles** i.e. old and new is always done.
- **Result of both practices are shown**
- It is **very effective in adoption**
- It is **conducted by a farmer under direct supervision** of the extension worker.
- It is used to show the **superiority of practice.**

c. National or composite demonstration

- It is a **composite of both method and result demonstration**

- It is **first line demonstration or front-line extension project**
- It is **conducted by the researcher** on the farmer's field.
- **The main objective is to show how production can be increased per unit area and per unit time.**
- Multiple cropping system are **usually followed and high yielding varieties** are used in this demonstration.
- **National demonstration was first time started in 1965 by ministry of food and agriculture**
- Now it has become a part of agricultural production programme

Training And Visit System

T and V system is a new extension. The concept of T and V system was evolved in 1973 by **Israeli extension expert Daniel benor** who was consultant on extension programme in world bank.

This system was first tried in seyhan irrigation project in turkey.

T and V system has two stages – i.e training and visit

1. Training- Training is imparted for transfer **of technical know whom from subject matter specialist (SMS) to extension worker.**

2. Visit- Extension worker has to visit to the farmers for transfer of know how obtained at training.

- It is **a govt sponsored** rural development programme
- It is also **called as a reformed extension or intensive agricultural extension**
- **T and V system is introduced in India in 1974 in Rajasthan canal and Chambal command area in MP to improve the effectiveness** of agricultural extension.
- **Under T and V Programme extension workers** are imparted training of the latest agricultural technology by the agriculture experts and the **trained worker after visiting the village diffuse the obtained know how to the selected farmers and instigate them to adopt it.** Besides the problem of the farmers are brought to the agriculture expert during the next training where the solution needs to be found.

Approaches For Organising Agricultural Extension Services

Axinn has identified the following eight models of or approaches to organising agricultural extension:

1. **The general agricultural extension approach:** It is the **common model** followed by most of the countries. It follows a **top-down transfer of technology approach** with the assumption **that communicating the details of improved technology** available with research institutes to farmers will result in increased productivity and production.
2. **The commodity specialised approach:** It focuses on a **particular crop or commodity such as coffee, rubber, sugarcane, etc.** It is **highly specialised in nature** with the planning and implementation of the programme controlled by the commodity organization.
3. **The training and visit approach:** It became popular during mid-1970s. This approach emphasises **regular training of extension workers on the latest techniques** and their visits to selected contact farmers. The programme planning and implementation is centrally controlled.
4. **The participatory extension approach:** It emphasises **formation of farmers' organisations and groups such as self-help groups and the need for learning from farmers' wisdom.** Farmers are involved not only in decision making related to programme planning and implementation, but also in controlling it locally.
5. **The project approach:** Here, the extension efforts are **concentrated on a specific area for a defined period of time along with a greater level of the infusion of resources** from outside. The planning and implementation of the programme is done by the project staff.
6. **The farming system approach:** The **main focus of this approach is to develop appropriate technology for the farmers in resource poor areas on a participatory basis.** The technologies are developed on the basis of holistic agro-ecosystem involving researchers, extension personnel, farm women and men.

7. The cost-sharing approach: Here, a part of **the cost of agricultural extension is paid by the farmers. Planning and implementation of the** programme is jointly shared by the implementing agency and the local people.

8. The educational institution approach: Here, the local agricultural research or educational institutions get involved in **planning and implementation of extension programmes on a limited scale** for the purpose of testing and transferring the new technologies developed by them.

Agricultural Extension in India after Independence:

Nilokheri Project:

Nilokheri Project **was started by S.K. Dey design to rehabilitate about 7000 displaced persons (immigrants) from Pakistan after partition.** He began this project using 100 acre of swampy land spreading in the **midst of Karnal and Kurukshetra.** The colony had its own dairy, poultry, piggery, printing press, engineering workshop, bone meal factory all run on cooperative lines. **The Scheme was called “Mazdoor Manzil”** because it was based on the principle of **‘he who would not work neither shall he eat.**

Etawah Pilot Project:

Under the leadership of Lt. Col. Albert Mayor of USA , the office of Etawah Pilot Project was established by **the U.P Govt. in October 1948 at Mahewa in the Etawah District.** **Initially 64 villages were selected around Mahewa District** for the development. It was also called Average District Plan because the project was initiated in the normal environment. **This programme named as pilot project means the work is for specific location and further it shows the path. This programme was the fore runner of CDP (Community Development Programme).**

Community Development Programme (CDP):

The Community Development Programme is a multipurpose extension programme which includes farm and **non-farm sectors of livelihood.** It is launched on 2nd October 1952. It achieved tremendous **results and benefits in the field of agriculture, rural development, social and economic development.**

15 Pilot Projects were started in 1952 in selected states with the financial assistance received from the Ford Foundation.

The project area was **divided into three development blocks, each comprising 100 villages and a population of 60,000 to 70,000.** The development blocks, in turn, **were divided into groups of 5-10 villages**, each group being in the charge of a multipurpose village-level worker.

The main aims of these projects were:

- to **increase agricultural production** by all possible means,
- to **tackle the problems of unemployment**,
- to **improve village** communications,
- to **foster primary education**, public health and recreation,
- to **improve housing**,
- to **promote indigenous handicrafts** and small-scale industries
- to **improve the villager's** lot through their own primary effort.

National Extension Service:

The **National Extension Service** programme was formulated in **April 1953** and it was inaugurated one year after **the 55 community projects that is, on October 2, 1953.**

It was a major development in the sphere of rural reconstruction in India. Since the basic idea underlying both the **community development** and National Extension Service programme was the same, the two were integrated under one agency at the Centre as well as in the states. Both the programmes were complementary and interwoven and ran concurrently. **The idea behind the National Extension Service Programme was to cover the entire country within a period of about 10 years.**

Intensive Agricultural District Programme (IADP)

There was fall in production **during 1957-58 due to drought increased the import of food grains.** Under these circumstances the agricultural scene of India was critically examined by the Government with the help of experts and Ford Foundation. **As a result, Intensive Agricultural District Programme (IADP) was launched during 1961.** The

IADP was started in several selected districts on a pilot basis and subsequently extended to nine more districts.

This programme was aimed at integrated and intensive approach to solve the problems of agricultural production through adoption of package of improved practices. The achievements accomplished through this programme were tremendous and it really made a dent into the rural poverty for its eradication through agriculture. This programme was popularly known as Package Programme.

Intensive Agricultural Area Program (IAAP):

To meet the demand of food for the vast population of the country, it was decided that at least 20 to 25 per cent of cultivated area of the country should be earmarked and selected for intensive agricultural development. Accordingly in 1964, IAAP came into operation in 114 districts of the country. **The IAAP's main objective was to bring about the progressive increase in production of main crops in selected areas by intensive and coordinated use of various aids to production.** The IAAP partially achieved its prime objectives by increasing food production in the selected area and on the selected crops.

High Yielding Varieties Programme (HYVP)

The IADP and IAAP were concerned with the package approach and intensive agriculture and these **programmes increased the food production to some extent. But in the later period, when yield production was stabilized, it could not meet the demand of the need of further food production and this experience directed the need of HYVP.** The **HYVP was launched in 1966** with a sole objective to increase the total food production by utilizing high yielding seeds of selected crops. The selected crops were paddy, wheat, bajra, jowar and maize.

Drought Prone Area Programme (DPAP)

In India, approximately, **19 per cent of the total land is frequently affected by drought.** To overcome this problem Government of India launched the DPAP during 5th Five Year Plan in June 1973. **The basic objective of DPAP was to reduce the severity of the**

impact of the drought and stabilizing the income of the people, particularly, the weaker sections of the society together with restoration of ecological balance.

Special emphases of DPAP were:

- **Development and management of irrigation** resources
- **Soil moisture conservation and afforestation**
- **Restructuring of cropping pattern** and pasture development
- **Changes in agronomic practices**
- **Livestock development of small farmers**, marginal farmers and agricultural labourers through special action plan.

Command Area Development Authority (CADA)

Government of India established Area Development Authority in July 1973 and later it was converted into CADA from June 1974. The basic concept of this CADA was coordination among the different departments like irrigation, soil conservation, agricultural extension and co-operation & credit agencies.

The basic aim of CADA was to bridge the gap between the irrigation potentialities and its actual coverage in irrigation of the land.

The CADA's more important function was to utilize the water available through its various irrigation projects and to distribute the same under command area through modernized distribution system.

Training and Visit (T & V) System of Agricultural Extension

The Training and Visit (T&V) system, the brain child of Dr Daniel Benor (World Bank Consultant, an Israeli extension expert) was introduced in India in 1974 for all round development of agricultural extension system in the country to introduce observation, **training and technology transfer to the farmers and extension workers** so as to enable them achieving greater productivity and production in the agricultural sector. **A basic feature of the T & V system was a systematic programme of in-service training of Village Extension Workers (VEWs)** and planned schedule of visits to the farmers' fields. The VEW was given training once in a fortnight on specific agriculture practices directly

related to farm operations scheduled for the coming weeks. There was concentration of efforts on important crops and agricultural practices.

In India this system was been **introduced first in 1974 in two states viz., Rajasthan and West Bengal with the help of World Bank and then spread to another 16 states** of the country aimed at building a professional extension service that is capable of **assisting farmers in raising production** and increasing incomes and of providing appropriate support for agricultural development.

Small Farmers' Development Agency (SFDA) and Marginal Farmers and Agricultural Labourers (MFAL)

SFDA started with the objectives to help and identify the special problems of small farmers as producers and also identifying marginal farmers and agricultural labourers, so that necessary measures can be taken to boost their occupation and helping them to get appropriate income as wages for support to their cultivation.

SFDA was first introduced in 168 district level agencies covering 1818 blocks. Later MFAL was combined to make this as a composite SF/MFAL blocks.

The major achievements this programme brought were that all small farmers, marginal farmers and agricultural labourers came in one umbrella to help them, identify their problems and solve them with the support from Government. It also helped agricultural labourers to get their minimum wages from the land owners where they used to put their labour

1978: IRDP (Integrated Rural Development Programme) By Mord

- **Beneficiaries** -Small and marginal farmers, agricultural labourers and rural artisans.
- District Rural Development Agencies carried out the program (DRDAs).
- **On 1st April, 1999, the IRDP and allied programmes** were merged into a single programme known as Swarnajayanti Gram Swarozgar Yojana (SGSY)
- **TRYSEM and DWCRA were two special components of IRDP**

1. 15 August, 1979: TRYSEM (Training of Rural Youth for Self-Employment) by MoRD

AIM - To provide **training facilities & create self-employment among the rural unemployed youths from families below poverty line (BPL)**

Its' main features were:

- To **provide need-based technical skills to rural youth aged 18-35 years** who belong to the category of the poor, to enable them to take up self-employment and to some extent even wage employment.
- **Training is provided at the regular training institutions** like polytechnics or under selected master craftsman.
- At **least 30 per cent of the trainees should belong to SCs/STs and at least 33 per cent should be women.**
- Marketing facilities for the products made by beneficiaries are improved.

2. 1982: DWCRA (Development of Women and Children in Rural Areas) by MoRD

On a pilot basis in 50 districts and extended to all the districts of the country.

AIM -Improve the **status and quality of life of poor women** and children in **rural** areas

- **Enable women to improve their earning capacity.**
- It operated in **conjunction with IRDP and TRYSEM**

1989: JRY (Jawahar Rozgar Yojana) by MoRD

By merging NREP & RLEGP

AIM - Assured **rozgar/ employment to ≥ 1 member of every BPL family** in the rural areas for **50 to 100 days in a year**

- **To generate meaningful employment opportunities** for the unemployed and underemployed in rural areas through the creation of economic infrastructure and community and social assets.
- **The schemes included construction of all-weather roads, minor irrigation works, soil and water conservation work, water harvesting structures, wasteland development, farm forestry etc.**

Overall objective:

- **To provide productive assets to the rural poor to enable them to generate additional or incremental income** out of the provided assets and also leave a surplus to repay the loan.

- **To provide employment to the rural poor** by creating durable community assets or undertaking other work programme both during busy and lean period.
- **To provide basic infrastructural facilities like clean drinking water**, medical facilities, education, electrification, rural roads etc.
- **To provide clinical and other types of guidance** and support to the unemployed to make them the potential entrepreneurs for self-employment.

Beneficiary –

- Primarily for **BPL families** in rural areas
- Preference was given to the **SCs, STs and the freed bonded labourers**.
- **30% reservation for women**

Revamped as new scheme Jawahar Gram Samridhi Yojana wef 1999.

Swarnajayanti Gramin Swarojgar Yojana

1 April, 1999: SGSY (Swarnajayanti Gramin Swarojgar Yojana): By MoRD

Self-Employment opportunities for rural poor → Low Income families (Swarojgaris)

By restructuring existing schemes, namely:

1. Integrated Rural Development Programme (IRDP)
2. Training of Rural Youth for Self-Employment (TRYSEM)
3. Development of Women and Children in Rural Areas (DWCRA)
4. Ganga Kalyan Yojana (GKY)
5. Million Wells Scheme (MWS)
6. **(Supply of Improved Toolkits to Rural Artisans) SITRA [1992]**
 - A variety of **artisans were supplied with kit** of improved hand-tools.
 - **Artisans had to pay 10%** & rest was subsidy from the Central Government

AIM - Establishing a large number of micro enterprises in the rural areas.

Beneficiaries - The list of BPL households identified through BPL census duly approved by Gram Sabha will form basis for identification of families for assistance under SGSY.

- **Rural poor such as those with land, landless labour**, educated unemployed, rural artisans and disable are covered under the scheme
- **SGSY will focus on vulnerable section of the rural poor:**
 - ❖ **SC/ST → 50%**

❖ **Women → 40%**

❖ **Disabled → 3%**

Funding Pattern - 75:25

MGNREGA The Mahatma Gandhi National Rural Employment Guarantee Act (MGNREGA) 2005

The Act was **notified in 200 rural districts** in its first phase of implementation (with **effect from Feb 2, 2006**). In **financial year 2007-08**, it was extended to additional 130 districts. The remaining districts were notified under MGNREGA with effect from April 1, 2008.

The mandate of the act is to provide 100 days of guaranteed wage employment in a financial year (FY) to every rural household whose adult members volunteer to do unskilled manual work. In this **Act "Adult"** means a person who has completed his **eighteenth year of age** and a "household" is defined as members of a family related to each other by blood, marriage or adoption and normally residing together and sharing meals.

Extension Role Of Agricultural Universities And Icar Institutes

The Indian Council of Agricultural Research (ICAR) is also involved in agricultural extension apart from its major role of research and education. The ICAR has contributed to agricultural extension by way of innovative extension approaches. In fact, the concept of and the project on National Demonstration, which was one of the key instruments for bringing about '**green revolution**' in India, was developed and launched by ICAR during the 1960s. The extension projects like Operational Research Project (ORP) and Krishi Vigyan Kendra (KVK) and Lab to Land Programme (LLP) were also introduced by ICAR. **The present thrust areas of extension activities of ICAR are related to technology assessment and refinement for identifying the location specific technology**, vocational training of farmers, training of extension personnel to update their **knowledge**, and providing **single window delivery system of technology products, services and information**.

Krishi Vigyan Kendras (KVKs)

Based on the recommendations of the Education Commission (1964-66), review by the Planning Commission, **inter-Ministerial Committee and other further recommendation by the committee headed by Dr. Mohan Singh Mehta appointed by ICAR in 1973,** the idea of establishment of Farm Science Centre (Krishi Vigyan Kendra) was evolved.

KVK, is an integral part of the National Agricultural Research System (NARS), aims at assessment of location specific technology modules in agriculture and allied enterprises, through technology assessment, refinement and demonstrations. KVKs have been functioning as Knowledge and Resource Centre of agricultural technology supporting initiatives of public, private and voluntary sector for improving the agricultural economy of the district and are linking the NARS with extension system and farmers.

The KVK scheme is 100% financed by Govt. of India and the KVKs are sanctioned to Agricultural Universities, ICAR institutes, related Government Departments and Non-Government Organizations (NGOs) working in Agriculture.

The first KVK, on a pilot basis, was established in 1974 at Pondicherry under the administrative control of Tamil Nadu Agricultural University, Coimbatore.

The mandate of KVK is Technology Assessment and Demonstration for its Application and Capacity Development.

To implement the mandate effectively, the following activities are envisaged for each KVK:

- 1. On-farm testing to assess the location specificity** of agricultural technologies under various farming systems.
- 2. Frontline demonstrations to establish production potential** of technologies on the farmers' fields.
- 3. Capacity development of farmers and extension personnel** to update their knowledge and skills on modern agricultural technologies

4. To **work as Knowledge and Resource Centre** of agricultural technologies for supporting initiatives of public, private and voluntary sector in improving the agricultural economy of the district.
5. **Provide farm advisories using ICT and other media** means on varied subjects of interest to farmers

Activities of Krishi Vigyan Kendras (KVKs):

1. **Planning and conducting survey of the operational area** in order to prepare the resource inventory with special reference to identifying the training needs of the farming community.
2. Planning and conducting production- **oriented, need-based short and long duration training courses both on campus as well as in the villages** for various target groups with priority on the weaker and the poor.
3. Developing and organizing **non-formal educational programmes by way of field days, farm visits, farmers fair, radio talk, Farm Science clubs** etc. as the follow up information support to training courses.
4. **Organizing farm science clubs, both in rural schools and in villages in order to induce in younger generation a liking for and an interest for agricultural** and allied sciences and scientific farming through supervised projects.
5. **Developing and maintaining the campus farms and demonstration units on scientific lines as the facilities for providing work experience** to the trainees as also disseminating the latest technical know-how.
6. **Providing practical facilities of the Kendra to the teachers** and the students of the vocational agriculture of the higher secondary schools.
7. Imparting **some general education to rural illiterates and school drop-outs** in order to make them not only good farmers but also better citizens.
8. **Providing added training facilities in the areas for home making** and nutrition education for rural community.
9. **Gradually enlarging the training facilities to encompass other important areas such as home crafts, cottage industries etc.** consistent to the requirements of the Integrated rural Development in collaboration with concerned organization.
10. **Implementing all such schemes of the ICAR and** other related organizations which intend to strengthen the training programmes of the Kendra.

The Sub Mission on Agricultural Extension (SMAE)

Sub Mission on Agricultural Extension under Green Revolution – is being Krishonnati Yojana implemented with an objective to restructure and strengthen the agricultural extension machinery with a judicious mix of extensive physical outreach of personnel, enhancement in quality through domain experts & regular capacity building, interactive methods of information dissemination, Public Private Partnership, pervasive & innovative use of Information & Communication Technology (ICT) / Mass Media, Federation of groups and convergence of extension related efforts under various Schemes and programmes of Government of India and the State Governments.

Support To State Extension Programmes for Extension Reforms Scheme

Support To State Extension Programmes for Extension Reforms Scheme ATMA SCHEME

In 1998, the Indian Government, with the support of World Bank, introduced the **Agriculture Technology Management Agency (ATMA)** under the **Innovation in Technology Dissemination (ITD)** component of the **National Agricultural Technology Project (NATP)**.

It was first introduced in 28 districts in seven states from 1998 to 2003 under the guidance of **MANAGE (National Institute of Agricultural Extension Management)**, an institution promoted by Ministry of Agriculture, Government of India. It was later expanded throughout the country in 2005

It was revamped, expanded and strengthened comprehensively in the year 2010 & 2014. Coverage of the scheme was increased in a phased manner.

The Scheme '**Support to State Extension Programmes for Extension Reforms (ATMA)**' implemented since 2005 has now been included as a Centrally Sponsored component of the Sub-Mission on Agriculture Extension (SMAE) under Krishonnati Yojana.

Presently, it is **under implementation in 704 districts of 28 States & 5 UTs of the country.**

This Scheme shall focus on the following key extension reforms:

- **Encouraging multi-agency extension** strategies involving Public/ Private Extension Service Providers.
- **Ensuring an integrated, broad-based extension delivery** mechanism consistent with farming system approach with a focus on bottom-up planning process.
- **Adopting group approach to extension in line with the identified needs and requirements of the farmers in the form of CIGs & FIGs** and consolidate them as Farmers Producer Organizations;
- **Facilitating convergence of farmer centric programmes** in planning, execution and implementation.
- **Addressing gender concerns by mobilizing farm women** into groups and providing training to them

National Mission on Agricultural Extension and Technology (NMAET)

National Mission on Agricultural Extension and Technology (NMAET):

In 2014, the **Cabinet Committee on Economic Affairs** has approved the **implementation of the National Mission on Agricultural Extension and Technology (NMAET) during the 12th Plan period.** The extension of NMAET and its components will be expanded and up-scaled appropriately and implemented in a more coordinated and convergent manner.

NMAET consists of 4 Sub Missions:

- Sub Mission on Agricultural Extension (**SMAE**)
- Sub-Mission on Seed and Planting Material (**SMSP**)
- Sub Mission on Agricultural Mechanization (**SMAM**)
- Sub Mission on Plant Protection and Plant Quarantine (**SMPP**)

The aim of the Mission was to restructure and strengthen agricultural extension to enable delivery of appropriate technology and improved agronomic practices to farmers. This was envisaged to be achieved by a judicious mix of extensive physical outreach and interactive methods of information dissemination, use of **ICT, popularisation of modern and appropriate technologies, capacity building and institution strengthening to promote mechanisation, availability of quality seeds, plant protection etc. and encourage aggregation of Farmers into Interest Groups (FIGs) to form Farmer Producer Organisations (FPOs).**

Information & Communication Technology (ICT) led extension services

1. **Mkisan-Use Of Basic Mobile Telephony**

Since the penetration of **smart phones with internet in rural areas is around 18% only, mobile telephony is considered to be the alternate** and the best option to deliver services to the farmers. Mobile telephony has transformed the tenor of our lives. The Department has developed a portal-mkisan (mkisan.gov.in), where, around **5.3 crore farmers are registered and experts/ scientists of different departments like IMD, ICAR, State Government, State Agriculture Universities send information to farmers in 12 local languages on a regular basis.** Information related to the weather such as likelihood of rainfall, temperature, etc. enables farmers to make informed decision in choice of seed varieties and decide on timing of sowing and harvesting. **With market information, farmers are better informed to sell produce, prevailing market prices and quantity demanded in the market.** Thus, they can make informed decisions to sell produce at the right price and at the right time. This helps in reducing distress sales by farmers due to market supply fluctuations.

2. **Farmers' Portal (www.farmer.gov.in)**

Farmers' Portal is a one stop shop for farmers where a farmer can get relevant information on a range of **topics including seeds, fertilizer, pesticides, credit, good practices, dealer network, availability of inputs, agromet advisory etc.** This information can be drilled down through the pictorial view of the Map of India placed on the Home page as well. **This centralized repository is the back bone of all mobile apps and SMS**

advisories. This portal provides information across all stages of crop management right from sowing of seeds till post harvesting.

An important **feature of this web-based portal** is that one can drill down to the block level and get information of the particular block.

3. Kisan Suvidha

It is a **Smart App for farmers, which is platform independent as well as available on Google Play Store.** All the **services/information relevant to farmers are linked here as a common platform.** It will include various schemes and services of Central and State Governments for Farmers. It includes Agriculture as well as Animal Husbandry Sector.

4. Agri Stack

Agri Stack project is an ambitious digital agriculture initiative of the Department of Agriculture and Farmers Welfare (DAFW), Ministry of Agriculture & Farmers Welfare, Government of India. Agri Stack is a collection of databases, registries, frameworks, standards, data schema, APIs, IT systems, that open up data related to agriculture, through interoperable systems for the development of innovative services and solutions by government entities and private sector. **Agri Stack, akin to Aadhaar, is envisioned to be the core data infrastructure for Agriculture that** will digitally enable better public and private ecosystems for service delivery across the agriculture value-chain. Using Agri Stack various government departments, and public and private sector entities can build a variety of use-cases for the benefit of farmers and farmer collectives, primarily.

5. Kisan call Centres

- **The Department of Agriculture & Cooperation (DAC), Ministry of Agriculture, Govt. of India launched Kisan Call Centres on January 21, 2004** across the country to deliver extension services to the farming community.
- The purpose **of these call centres is to respond to issues raised by farmers, instantly, in the local language.** There are call centres for every state which are expected to handle traffic from any part of the country. Queries related to agriculture and allied sectors are being addressed through these call centres.

6. Agri Clinics & Agribusiness Centres

- The **Ministry of Agriculture and farmers welfare, Government of India, in association with NABARD has launched in 2002**, a unique programme to take better methods of farming to each and every farmer across the country.
- Irrespective of whether you are a fresh graduate or not, or whether you **This programme aims to tap the expertise available in the large pool of Agriculture Graduates.** are currently employed or not, you can set up your own Agri Clinic or Agribusiness Centre and offer professional extension services to innumerable farmers. Committed to this programme, **the Government is now also providing start-up training to graduates in Agriculture, or any subject allied to Agriculture like Horticulture, Sericulture, Veterinary Sciences, Forestry, Dairy, Poultry Farming, and Fisheries, etc.** Those completing the training can apply for special start-up loans for venture.

Last year Previous year Question 2023

1. What is agriculture extension education? Explain 12 tools used by extension workers for teaching farmers.